

# **MUTUAL COUPLING REDUCTION IN MIMO PATCH ANTENNA ARRAY FOR Wi-Fi AND BLUETOOTH APPLICATIONS**

A Project Report

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under the supervision of

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**Certificate**

This is to certify that this project report titled “**Mutual coupling reduction in MIMO Patch Antenna Array for Wi-Fi and Bluetooth applications**”, submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Electronics and Communication Engineering, by **J Rohith, Peddakudala Yashwanth, C Tej Siddanth**, is a bonafide record of work carried out by them, under my supervision and that it has not been submitted, to the best of my knowledge, in part or in full, for the award of any other degree or diploma.

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We do hereby declare that this project report titled “**Mutual coupling reduction in MIMO Patch Antenna Array for Wi-Fi and Bluetooth applications**”, submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Electronics and Communication Engineering**, is a true record of work carried out by us, under the supervision of **Ms. Mekaladevi V** and that all information contained herein, which do not arise directly from our work, have been properly acknowledged and cited, using acceptable international standards. Further, we declare that the contents of this report have not been submitted, in part or in full, for the award of any other degree or diploma.

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## **Abstract**

With the growing use of Wi-Fi and Bluetooth in everyday devices, there is a constant need for compact antennas that can still deliver good performance. When patch antenna elements are placed close to each other, especially within half a wavelength, mutual coupling becomes a serious issue. This reduces isolation and affects efficiency and the overall quality of communication.

In this project, a  $1 \times 2$  patch antenna array is designed specifically for Wi-Fi and Bluetooth applications. To deal with the coupling between the antenna elements, a decoupling structure is added to the design. The aim is to achieve more than 20 dB isolation while keeping the array compact. Simulations are carried out in Ansys HFSS, which is used to fine-tune the geometry, reduce return loss, and ensure that the antenna maintains a broadside radiation pattern. The array is also optimised to produce a gain of around 4 dB, making it suitable for reliable wireless communication.

Once the design is finalised through simulations, a prototype will be fabricated and tested. The measured values, such as S-parameters and radiation patterns, will then be compared with the simulation results to confirm the performance. The proposed antenna array is expected to provide a simple and effective solution for Wi-Fi and Bluetooth applications, offering high isolation in a compact form without losing radiation stability.

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## **List of Abbreviations**

|           |                                                       |
|-----------|-------------------------------------------------------|
| ANSYS     | Analysis System                                       |
| CMA       | Characteristic Mode Analysis                          |
| CP        | Circular Polarization / Circularly Polarized          |
| ECC       | Envelope Correlation Coefficient                      |
| FR        | Flame Retardant                                       |
| FSS       | Frequency Selective Surface                           |
| HEM       | Hybrid Electric Magnetic Mode (dielectric patch mode) |
| HFSS      | High Frequency Structure Simulator                    |
| IFA       | Inverted F-Antenna                                    |
| ISM       | Industrial, Scientific, and Medical (band)            |
| MCM       | Mode Cancellation Method                              |
| MIMO      | Multiple Input Multiple Output                        |
| PCB       | Printed Circuit Board                                 |
| PDSS      | Parasitic Dollar-Shaped Structure                     |
| RT/Duroid | Rogers Teflon-based Duroid                            |
| SMA       | Sub Miniature version A (connector type)              |
| VNA       | Vector Network Analyser                               |
| Wi-Fi     | Wireless Fidelity                                     |

## List of Symbols

| Symbol              | Meaning                                                    |
|---------------------|------------------------------------------------------------|
| $f$                 | Frequency                                                  |
| $\varepsilon_r$     | Relative permittivity of the substrate                     |
| $\varepsilon_{eff}$ | Effective dielectric constant                              |
| $L$                 | Length of the patch                                        |
| $W$                 | Width of the patch                                         |
| $h$                 | Height of the substrate                                    |
| $\Delta L$          | Extension length due to fringing                           |
| $L_{eff}$           | Effective length of the patch                              |
| $c$                 | Velocity of light                                          |
| $\lambda$           | Wavelength                                                 |
| $f_r$               | Resonant frequency                                         |
| $ECC$               | Envelope Correlation Coefficient (MIMO performance metric) |



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# 1. Introduction

With the rapid growth of modern wireless communication systems, there is a strong demand for antennas that are compact, efficient, and capable of providing reliable connectivity at higher data rates. One of the key technologies that helps achieve this is Multiple Input Multiple Output (MIMO), which uses multiple antennas at both the transmitter and receiver to improve link quality and data throughput. However, when several antenna elements are placed close to each other, **mutual coupling** occurs, which degrades the overall system performance. Mutual coupling causes unwanted interaction between the antenna elements through electromagnetic fields. This leads to problems such as reduced isolation, poor impedance matching, and lower radiation efficiency. It can also increase the correlation between channels, which affects the diversity and capacity of the system. Therefore, minimising mutual coupling is a major challenge in designing efficient MIMO antenna arrays. In this project, a **MIMO microstrip patch antenna array operating at 2.4 GHz** is to be designed and analysed. The antenna is targeted for wireless communication applications such as **Wi-Fi, Bluetooth devices**, where compact size and high performance are essential. Simulation and optimisation are carried out using full-wave electromagnetic simulation software, followed by fabrication and testing to verify the results. The outcomes are expected to contribute toward the development of efficient antenna systems for wireless communication applications.

## 1.1 Motivation

In today's world, almost every modern device — from smartphones and smartwatches to routers and IoT modules — relies heavily on wireless connectivity. As these technologies evolve, the demand for faster data rates and more reliable connections continues to grow. Multiple-Input Multiple-Output (MIMO) antenna systems have become a key solution to meet these needs by improving throughput and link quality without increasing bandwidth or transmit power. However, one of the major challenges faced by MIMO systems is mutual coupling between antenna elements, which can degrade overall system performance. This issue particularly becomes critical when antennas are placed close together in compact devices, as in most Wi-Fi and Bluetooth modules operating at 2.4 GHz.

The motivation for this project stems from the need to design antennas that can maintain high isolation while being compact and efficient. Since portable devices have

limited space, keeping a sufficient distance between antenna elements is not always feasible. Therefore, developing effective decoupling techniques can significantly enhance the performance of MIMO systems. Understanding and reducing mutual coupling not only improves isolation and gain but also helps achieve a more stable radiation pattern and better signal integrity — all of which are crucial for practical wireless communication systems.

Ultimately, this project motivates us to contribute, even in a small way, to ongoing research in modern communication systems. By experimenting with decoupling structures and validating them through simulation and measurement, we aim to develop a simple yet effective design that can inspire future innovations in compact antenna systems. The satisfaction of designing, fabricating, and testing something that can actually improve wireless communication performance makes this project truly rewarding for us.

## **1.2 Challenges**

One of the main challenges in designing a MIMO patch antenna array is achieving high isolation between antenna elements without increasing the overall size of the structure. Since most Wi-Fi and Bluetooth devices operate in compact enclosures, the antenna elements are usually placed very close to each other — often less than half the wavelength apart. This proximity leads to strong mutual coupling, which distorts the radiation pattern, reduces gain, and affects impedance matching. Finding a design that maintains performance while keeping the antenna compact is not straightforward, and it requires multiple iterations in simulation and fine-tuning of parameters..

Another major difficulty lies in selecting and optimising the right decoupling structure. There are many techniques like slots, parasitic elements, and defected ground structures, each with its own pros and cons. Choosing the most effective one for 2.4 GHz operation involves understanding how the surface currents interact between elements and how each modification impacts bandwidth, isolation, and gain. Sometimes, even a small change in geometry or substrate parameters can shift the resonant frequency or degrade other characteristics, which makes the design process quite sensitive and time-consuming.

Simulating and validating the design using software like Ansys HFSS also poses challenges. Accurate modelling requires proper meshing, boundary setup, and excitation, which are easy to get wrong for beginners. Achieving convergence in results and interpreting the S-parameters correctly takes patience and practice. Additionally, once the simulation results are obtained, correlating them with real-world measurements can be difficult due to fabrication tolerances and losses in the substrate and connectors.

Finally, practical implementation adds its own hurdles. Fabricating the antenna with precise dimensions, ensuring good soldering for SMA connectors, and testing it in the anechoic chamber using a VNA demands both technical skill and attention to detail. Even minor errors during fabrication can cause deviations in performance compared to the simulated model. Overcoming these challenges is part of what makes the project both tough and rewarding. It teaches us the importance of accuracy, iteration, and hands-on experimentation in real engineering work.

## 2. Literature Survey

In the paper *Compact Self-Isolated Four-Element MIMO Antenna for WLAN and ISM Bands*, the focus was on a four-element MIMO antenna operating in the 2.25–2.87 GHz range, which covers the WLAN and ISM 2.4 GHz bands. The antenna is based on printed dipole elements with integrated  $\lambda/4$  balun feeding, ensuring proper impedance matching between the coaxial line and the balanced dipole. What makes the design impressive is how it uses self-isolation through the smart arrangement of elements rather than relying solely on external decoupling networks. By placing dipoles orthogonally in a compact square arrangement, mutual coupling is naturally reduced, and then a simple printed decoupling element further suppresses residual coupling. The measured performance showed isolation better than 19 dB, ECC less than 0.0009, and a peak gain of 8.6 dBi, which is excellent for such a compact structure. For me as a student, this shows that effective antenna design is not always about adding complexity; sometimes, a thoughtful arrangement with the right feeding structure can achieve high isolation and efficiency while keeping the antenna simple and manufacturable.[1]

In the paper *Decoupling Between Extremely Closely Spaced Patch Antennas by Mode Cancellation Method*, focus is on microstrip patch antennas that were arranged with an extremely small edge-to-edge spacing of just  $0.016\lambda_0$  ( $\approx 2$  mm at 2.45 GHz). The patches were probe-fed and operated in the 2.39–2.53 GHz band. Because of such close spacing, the antennas originally showed very poor isolation (about 5 dB). To solve this, a mode cancellation method (MCM) was introduced, where a lumped inductance or a small inductive strip was inserted between the patches. This inductive path tuned the common-mode (CM) and differential-mode (DM) impedances to a similar state, which effectively cancelled the coupling. With this simple adjustment, the isolation improved dramatically to better than 15 dB, while maintaining stable radiation with a symmetric broadside beam and gain around 8.9 dBi. What stands out here is that a small feeding modification and a basic inductive element can fundamentally reshape how the fields interact. As a student, it taught me that sometimes the smallest, simplest additions, when guided by strong theory, can solve what looks like an impossible challenge in compact antenna design.[2]

In the paper *Mutual Coupling Reduction of a Dual-Band Four-Port Patch Antenna with Co-Polarised Radiation*, the study examined a dual-band four-port rectangular patch antenna designed to operate around 2.35 GHz and 3.59 GHz. All four ports were excited in a co-polarised configuration using a standard probe-feed. The challenge with such designs

is that using a single radiator with multiple ports usually results in very high mutual coupling. To overcome this, characteristic mode analysis (CMA) was applied, which guided the placement of shorting pins and ring slots on the patch. These structural changes altered the modal distribution of electric fields, effectively pushing field nulls closer to the unexcited ports. The result was a substantial reduction of mutual coupling to below  $-15$  dB across both bands, while still maintaining the advantages of a low-profile, single-layer, and single-element radiator. Importantly, the antenna preserved stable broadside beams and co-polarised radiation patterns, which are critical for practical communication systems. As a student, what I found inspiring is how advanced analysis tools like CMA can make antenna design more scientific rather than trial-and-error. By directly controlling how modes combine, engineers can achieve compact, dual-band, and multi-port antennas that are both efficient and reliable.[3]

In the paper *A Compact Multifrequency Broadband Dual-Antenna System With Mutual Coupling Reduction for Wi-Fi 6E Application*, the authors present a compact dual-antenna system designed specifically for Wi-Fi 6E applications, which supports multiple frequency bands including 2.4 GHz, 5 GHz, and the new 6 GHz band (5.925–7.125 GHz). The work focuses on reducing mutual coupling between two closely placed antennas in devices like laptops, where space is extremely limited. The proposed antenna system consists of two identical Inverted-F Antennas (IFAs) printed on an FR4 substrate of size  $6.5 \times 41$  mm<sup>2</sup>, separated by just 2 mm. To minimize coupling, the authors introduced three innovative decoupling techniques: A decoupling inductor (L1) placed between the open ends of the IFAs to handle low-frequency decoupling (2.4 GHz band), A modified decoupling branch (L2) designed to improve isolation in the higher 5–6 GHz region, A modified neutralization line (L3) that ensures phase cancellation of mutual currents at higher frequencies. The optimised inductance values were  $L1 = 3.6$  nH,  $L2 = 22$  nH, and  $L3 = 28$  nH, tuned using HFSS simulations. These components helped the antenna achieve  $|S_{21}| > 22$  dB isolation across all Wi-Fi 6E frequency bands, which is a remarkable improvement compared to traditional dual-antenna systems. In terms of radiation, the antenna achieved peak gains of 4.27 dBi (2.4 GHz) and 4.43 dBi (5–7 GHz), with an efficiency above 40%. The Envelope Correlation Coefficient (ECC) remained below 0.3, confirming good MIMO performance. The radiation pattern was almost omnidirectional, suitable for laptop integration. Key Technical Details: Frequency Bands: 2.4–2.484 GHz, 5.15–5.825 GHz, 5.925–7.125 GHz, Antenna Type: Dual Inverted-F Antenna (IFA), Feeding Technique: Microstrip feed on FR4 substrate ( $\epsilon_r = 4.4$ , thickness = 1 mm), Isolation Achieved:  $> 22$  dB,

Antenna Size:  $6.5 \times 41 \text{ mm}^2$ . Application: Wi-Fi 6E-enabled laptops and compact wireless systems.[4]

In paper Mutual Coupling Reduction in Dual-Band MIMO Antenna Using Parasitic Dollar-Shaped Structure for Modern Wireless Communication, focuses on a dual-band MIMO antenna developed for 5G Sub-6 GHz and Wi-Fi 6E applications, which require high-speed connectivity and low interference in compact devices. The major challenge addressed here is mutual coupling reduction between closely placed antenna elements. The proposed MIMO antenna uses two identical monopole radiators, each composed of a rectangular patch, an inverted-E-shaped arm, and an asymmetrical T-shaped strip. These elements work together to produce two resonant frequencies—2.5 GHz (for 5G Sub-6) and 4.5 GHz (for Wi-Fi 6E). The overall antenna size is just  $45 \times 30 \text{ mm}^2$ , with the two elements separated by 5 mm ( $\approx 0.04\lambda$  at 2.5 GHz). The novelty of this design lies in a Parasitic Dollar-Shaped Structure (PDSS) placed between the two monopoles. The PDSS is made up of two interlocking C-shaped loops and a vertical rectangular strip acting as a stop-band filter. This unique geometry suppresses surface waves, reducing mutual coupling to  $-15 \text{ dB}$  while keeping the impedance bandwidths wide (2.39–2.57 GHz and 3.82–6.95 GHz). The antenna employs a Coplanar Waveguide (CPW) feed for better impedance matching and simple fabrication on an FR4 substrate ( $\epsilon_r = 4.4$ , thickness = 0.8 mm). Measurements confirmed good agreement with simulations, showing an efficiency of 70–75%, a gain of 2.6 dBi, and  $\text{ECC} < 0.02$ , which meets MIMO performance standards. Key Technical Details: Frequency Bands: 2.39–2.57 GHz (5G Sub-6), 3.82–6.95 GHz (Wi-Fi 6E), Antenna Type: Dual-band monopole MIMO antenna, Feeding Technique: CPW (Coplanar Waveguide) feed, Decoupling Structure: Parasitic Dollar-Shaped Structure (PDSS), Isolation Achieved:  $-15 \text{ dB}$  across both bands, Antenna Size:  $45 \times 30 \text{ mm}^2$ , Performance: Gain  $\approx 2.6 \text{ dBi}$ , Efficiency  $\approx 70\%$ ,  $\text{ECC} < 0.02$ .[5]

In the paper Electric-Field Null Bending of a Single Dual-Port Patch Antenna for Colinear Polarisation Decoupling Using Characteristic Modes Analysis, this paper proposes a very interesting concept of electric-field null bending in a single dual-port microstrip patch antenna (MPA) to achieve colinear polarisation decoupling. The main motivation is to reduce mutual coupling in compact antennas, especially in in-band full-duplex (IBFD) communication systems where transmission and reception happen simultaneously at the same frequency. Traditionally, improving isolation between two ports of a microstrip antenna requires additional structures like defected ground structures (DGS),



electromagnetic bandgap (EBG) cells, or neutralisation lines. These methods increase complexity and size. However, this paper achieves excellent decoupling by cleverly combining two resonant modes of the same patch — mode 2 (TM<sub>01</sub>) and mode 4 (TM<sub>20</sub>) — to bend the electric-field nulls in such a way that each feed port lies near a null of the other, reducing the coupling naturally. To accomplish this, the authors use Characteristic Mode Analysis (CMA) to study and control how the modes interact. They then insert six shorting pins below the patch to bring the resonant frequencies of the two modes close together around 5.4 GHz. Further, small circular slots are cut near the feed ports to improve impedance matching. When tested, the proposed antenna operated efficiently between 5.30 and 5.54 GHz, maintaining  $|S_{11}| < -10$  dB and an isolation of  $|S_{12}| = -17$  dB, which is a significant improvement over the  $-3.5$  dB of a conventional dual-port MPA. The antenna preserved a broadside radiation pattern with colinear polarisation, making it ideal for IBFD and MIMO systems that require high isolation and same-polarisation transmission. The measured gain was around 5.0 dBi with good agreement between simulation and experiment. Key Technical Details: Operating Frequency: 5.30–5.54 GHz, Antenna Type: Single-layer, dual-port Microstrip Patch Antenna (MPA), Feeding Technique: Probe-fed microstrip ports (two symmetrical feed points), Isolation Achieved:  $-17$  dB, Peak Gain:  $\approx 5.0$  dBi, Structure Additions: Six shorting pins and circular slots for impedance matching, Applications: In-Band Full Duplex (IBFD), Military, MIMO systems [6]

In the paper Wideband Self-Decoupled Closely Placed Dielectric Patch Antenna Array Without Additional Structures, suggests a wideband dielectric patch antenna (DPA) array design where mutual coupling between elements is minimised without using any extra structures like split-ring resonators or metasurfaces. What I found interesting is that instead of adding external decoupling parts, they make use of the natural resonant modes of the dielectric patch itself — mainly the HEM<sub>11 $\delta$</sub>  mode and the TE<sub>12</sub> mode. By carefully choosing the dielectric constant, both modes can be excited and decoupled simultaneously. The design was first validated on a two-element array and later extended to a four-element array. The measured results showed an impedance bandwidth of about 10.5% (4.73–5.26 GHz) with an isolation better than 21.5 dB, which is pretty good considering the elements are closely spaced (about  $0.5\lambda_0$  apart). Moreover, the array supports beam scanning up to  $\pm 46^\circ$  in the E-plane, which makes it useful for MIMO systems in modern wireless communication. Key Technical Points: Frequency band: 4.7–5.3 GHz, Antenna type: Dielectric Patch Antenna (DPA), Feeding technique: Microstrip-coupled slot feed on a Rogers substrate, Performance: 10.5% bandwidth,  $>21$  dB isolation, scanning up to  $\pm 46^\circ$ . [7]

In the paper Self-Decoupled 8-Element Dual Polarised MIMO Antenna for B5G Handsets With Sub-6 GHz and mmWave Integration proposes a dual-band, 8-element self-decoupled MIMO antenna system designed for 5G and Beyond-5G (B5G) smartphones. What stood out to me is that the design cleverly uses F-shaped slots with a slotted ground plane to achieve two independent resonances — one in the sub-6 GHz band (4.7–4.9 GHz, n79) and another in the mmWave band (28–29 GHz, n257). Unlike many earlier works that require external decoupling structures (like metasurfaces or parasitic elements), this design achieves high isolation just by the orthogonal placement of elements. It's neat because they managed to keep the antenna compact while still handling both low-band and high-band requirements for future smartphones. The results showed isolation better than 14.4 dB at sub-6 GHz and >25 dB at mmWave, with peak gains of 4.9 dBi (n79) and 7.2 dBi (n257). The antenna efficiency was also impressive, around 90%, which is quite high considering the compact size and the fact that there are 8 elements squeezed into a phone-sized board. Key Technical Points: Frequency bands: Sub-6 GHz (4.7–4.9 GHz, n79) and mmWave (28–29 GHz, n257), Antenna type: 8-element dual-polarised MIMO antenna (slot-based), Feeding technique: Single-fed with F-shaped feeding strips on a RO4003C substrate (0.8 mm thick), Performance: High isolation (>14.4 dB / >25 dB), peak gain (4.9 dBi / 7.2 dBi), and ~90% efficiency.[8]

In the paper Mode-Counteraction Based Self-Decoupling in Circularly Polarised MIMO Microstrip Patch Array, a circularly polarised (CP) MIMO microstrip patch antenna (MPA) array that achieves self-decoupling without any extra decoupling structures. The authors made use of the mode counteraction effect. By exciting two orthogonal modes (TM<sub>10</sub> and TM<sub>01</sub>) within the rectangular patch, the antenna not only generates circular polarisation but also creates a null-field region in the adjacent patch. When the feed of the second patch is placed in this null-field region, it cannot pick up unwanted coupled signals, resulting in an extremely low mutual coupling of about –59.4 dB at 5.5 GHz. The design uses a diagonal probe feed to excite both modes properly. The fabricated prototype worked very close to the simulations, with only a small frequency shift (due to PCB tolerance). The results showed a 3 dB axial ratio bandwidth of ~2.15% (5.52–5.64 GHz), gain of ~4.8 dBi, and excellent port isolation. The authors also extended this approach to corner-truncated CP patch arrays and even a 1×4 MIMO array, which proved that the method is versatile and not limited to two-element arrays. Key Technical Points: Frequency band: ~5.45–5.63 GHz (centred at 5.5 GHz), Antenna type: Circularly Polarised Microstrip Patch Array (MIMO),

Feeding technique: Coaxial probe (diagonal-fed), Performance: Isolation as low as  $-59.4$  dB, 3 dB AR bandwidth  $\sim 2.15\%$ , gain  $\sim 4.8$  dBi.[9]

In the paper A Self-Decoupling Method for MIMO Antenna Array Using Characteristic Mode of Ground Plane, this paper introduces a self-decoupling method for MIMO antenna arrays, which eliminates the need for extra decoupling structures. The technique relies on the characteristic modes (CMs) of the ground plane that naturally create null-field regions. When one antenna excites this mode and another is positioned within the null field, high isolation (up to 57 dB) can be achieved. The method is versatile and suitable for different antenna types and array configurations. The core idea is to use the specific characteristic mode (Mode 2) of a rectangular ground plane, which shows two parallel null-field regions. Steps followed: Analyse the CMs of the ground plane using Altair Feko, identify the mode that generates parallel null regions (Mode 2 at 3.9–4 GHz), and excite this mode with one antenna. Place the second antenna's feed point in the null region.[10]

### **3. Methodology**

#### **3.1 Overview**

The design process began with the development of a single-element microstrip patch antenna using ANSYS HFSS simulation software. The antenna geometry, substrate type, and feeding mechanism were carefully selected to achieve resonance at the desired frequency band. Once the basic element was designed, its performance was analysed in terms of S-parameters and gain plots to ensure good impedance matching and radiation efficiency. Based on this single-element analysis, a  $1 \times 2$  MIMO antenna array was designed without any decoupling structure to study the mutual coupling behaviour between the antenna elements and understand its impact on system performance. After identifying coupling issues, a suitable decoupling structure was inserted into the MIMO configuration to enhance isolation and improve overall gain performance. The optimised antenna design was then fabricated using appropriate PCB materials and standard microstrip fabrication techniques. The fabricated prototype underwent experimental validation using a vector network analyser and anechoic chamber measurements to compare simulated and measured results. Finally, all observations, simulation data, and test outcomes were compiled and documented to evaluate the antenna's effectiveness and verify the design objectives.

#### **3.2 Software Setup**

The design and simulation work for the antenna were carried out using ANSYS HFSS (High-Frequency Structure Simulator), which is a 3D electromagnetic simulation tool widely used for antenna modelling and analysis. The setup process began by defining the operating frequency of 2.4 GHz, which lies within the ISM (Industrial, Scientific, and Medical) band commonly used for Wi-Fi and Bluetooth applications. The simulation environment was configured with appropriate solution frequency and frequency sweep range to ensure accurate results around this band.

A suitable substrate material (such as Rogers RT/Duroid) was selected within HFSS, and its dielectric constant, loss tangent, and thickness were specified. The ground plane, patch, and feedline were modelled according to the calculated dimensions for 2.4 GHz resonance. Radiation boundaries and wave ports were assigned to simulate realistic free-space conditions and signal excitation. The model was then solved using the HFSS setup wizard, and key parameters such as  $S_{11}$  (return loss), gain, and radiation pattern were analysed. This

setup provided a reliable simulation environment for both single-element and MIMO antenna configurations, helping to predict and optimise performance before fabrication.

Simulation setup:

Solution frequency: 2.4GHz, Substrate: Rogers RT duroid 5880 ( $\epsilon_r = 2.2$ ), Substrate height: 1.6mm.

## 4. Antenna Design

The antenna is designed to operate at 2.4 GHz, a frequency commonly used for wireless communication systems such as Wi-Fi, Bluetooth, and ISM band applications. A rectangular microstrip patch antenna configuration was chosen because of its compact size, ease of integration with circuits, and reliable directional radiation characteristics. For better performance, the design utilised Rogers RT/Duroid 5880 as the substrate material due to its low dielectric constant ( $\epsilon_r = 2.2$ ) and low loss tangent ( $\tan \delta = 0.0009$ ).

The return loss of the patch antenna is determined by the width  $W$ . The frequency of operation of the patch antenna is determined by the length  $L$ . The centre frequency is given by:

$$f = \frac{c}{2L\sqrt{\epsilon_r}} = \frac{1}{2L\sqrt{\epsilon_0\epsilon_r\mu_0}} \quad (4.1)$$

Width of the patch is given by :

$$W_p = \frac{c}{2f_0\sqrt{\frac{\epsilon_r+1}{2}}} \quad (4.2)$$

$$W = \frac{3 \times 10^8}{2(2.4 \times 10^9)} \sqrt{\frac{2}{2.2+1}} = 49.38 \text{ mm} \quad (4.3)$$

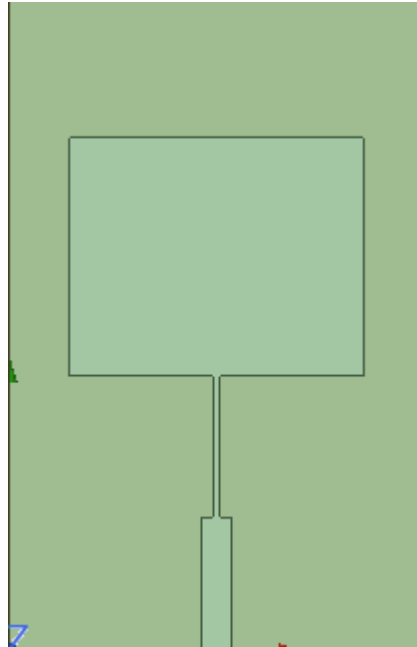


Figure 4.1- Simulated model of a unit element of an Array

Effective dielectric constant ( $\epsilon_{eff}$ ):

$$\epsilon_{eff} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \sqrt{\frac{1}{1 + 12 \frac{h}{W_p}}} \quad (4.4)$$

$$\frac{W}{h} = \frac{49.38}{1.6} = 30.86 \quad (4.5)$$

$$\epsilon_{eff} = 2.03 \quad (4.6)$$

The extension length:

$$\Delta L = 0.412h \frac{(\epsilon_{eff} + 0.3) \left( \frac{W_p}{h} + 0.264 \right)}{(\epsilon_{eff} - 0.258) \left( \frac{W_p}{h} + 0.8 \right)} \quad (4.7)$$

$$\Delta L = 0.74 \text{ mm} \quad (4.8)$$

Effective length ( $L_{eff}$ ):

$$L_{eff} = \frac{c}{2f_0 \sqrt{\epsilon_{eff}}} \quad (4.9)$$

$$L_{eff} = 30.2 \text{ mm} \quad (4.10)$$

Physical length of the patch is given by:

$$L = L_{eff} - 2\Delta L \quad (4.11)$$

$$L = 28.7 \text{ mm} \quad (4.12)$$

Impedance matching for quarter-wave transformer feeding:

$$Z_t = \sqrt{Z_0 \times Z_{in}} \quad (4.13)$$

$$Z_{in} = \frac{40(\epsilon_r)^2}{\epsilon_r - 1} \left( \frac{L}{W} \right)^2 \quad (4.14)$$

$$Z_0 = 50 \Omega, Z_{in} = 100 \Omega, Z_t = 70.7 \Omega$$

$$L_t = \frac{\lambda_g}{4} \quad (4.15)$$

$$\lambda_g = \frac{c}{f_r \sqrt{\epsilon_{eff}}} \quad (4.16)$$

$$L_t = 30.8 \text{ mm}$$

Here, the length is  $\lambda/4$  , and the width can be calculated as follows:

If  $\left(\frac{W}{H}\right) > 1$  :

$$Z_0 = \frac{120\pi}{\sqrt{\epsilon_{eff}}\left(\frac{W}{H} + 1.393 + \frac{2}{3} \ln\left(\frac{W}{H} + 1.444\right)\right)} \quad (4.17)$$

Using Equations 4.4 and 4.17 Width can be calculated.

If  $\left(\frac{W}{H}\right) < 1$  :

$$Z_0 = \frac{60}{\sqrt{\epsilon_{eff}}} \ln\left(8\left(\frac{H}{W}\right) + 0.25\left(\frac{W}{H}\right)\right) \quad (4.18)$$

Using Equations 4.4 and 4.18 Width can be calculated.

Table 4-1-Calculated Antenna Dimensions for single element

| Variables | Value         |
|-----------|---------------|
| $W$       | 49.38 mm      |
| $L$       | 28.7 mm       |
| $W_g$     | 70 mm         |
| $L_g$     | 70 mm         |
| $h$       | 1.6 mm        |
| $Z_0$     | 50 $\Omega$   |
| $Z_t$     | 70.7 $\Omega$ |
| $L_t$     | 30.8 mm       |
| $W_f$     | 4.9 mm        |
| $W_t$     | 1.09 mm       |



The antenna is built on a Rogers RT/duroid 5880 substrate, which has a low dielectric constant ( $\epsilon_r = 2.2$ ) and a low loss tangent, making it ideal for high-efficiency patch antennas. The substrate thickness ( $h$ ) is chosen as 1.6 mm.

The patch width ( $W$ ) is calculated based on the operating frequency and dielectric constant to control the resonant behaviour and radiation efficiency, resulting in a width of 49.8 mm. To account for the effect of fringing fields at the edges, an effective dielectric constant of 2.03 is used to compute the effective length of the patch. The length extension ( $\Delta L$ ), caused by fringing, is calculated to be 0.74 mm, which leads to an actual patch length ( $L$ ) of 28.7 mm.

A 50  $\Omega$  microstrip line is used to match standard transmission lines. To efficiently match the patch input resistance with the feedline, a quarter-wave transformer is inserted. Its characteristic impedance ( $Z_t$ ) is calculated as 70.7  $\Omega$ , and its length ( $L_t$ ) is a quarter of the guided wavelength ( $\approx 30.8$  mm) based on the effective dielectric constant. The corresponding microstrip widths are computed using standard transmission line equations, giving  $W_f = 4.9$  mm for the 50  $\Omega$  line and  $W_t = 1.09$  mm for the 70.7  $\Omega$  transformer line.

#### 4.1 Two-element Array

To improve the performance of the single patch antenna, particularly its gain and directivity, a two-element linear array is designed using the same rectangular microstrip patch on Rogers RT/duroid 5880. Essentially, two identical patches, each designed to resonate at 2.4 GHz, and arranged side by side along a straight line. The goal of this setup is to have the two elements work together to create a stronger, more focused radiation in the broadside direction while keeping the impedance properly matched. Each patch is fed using its own quarter-wave transformer, which connects to a 50  $\Omega$  feedline. This ensures that the patches receive the correct amount of power and remain well-matched to the source. The spacing between the centres of the two patches is chosen to be approximately  $0.5 \lambda_0$  at 2.4 GHz. This spacing is ideal because it helps the radiated waves from the two patches add constructively in the main lobe direction, enhancing the overall gain without creating unwanted grating lobes, also increasing isolation between the patches.

The patch dimensions, transformer parameters, and substrate properties remain exactly the same as the single patch design. The  $50\ \Omega$  feedline splits into two paths, each feeding one patch through its quarter-wave transformer. The transformers maintain proper impedance matching despite the two patches being connected, so the input sees a clean  $50\ \Omega$  load.

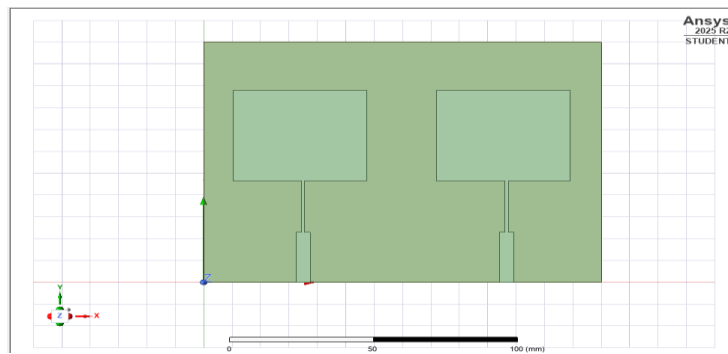


Figure 4.2-Two element array

## 5. Results and Discussion

### 5.1 Performance analysis of a single-element antenna

The designed rectangular microstrip patch antenna was analysed to assess its performance at the target frequency of 2.4 GHz, which is part of the ISM band. From a theoretical perspective, a microstrip patch resonates when the length of the patch corresponds to approximately half a wavelength in the dielectric medium. In our design, the effective length was calculated by considering the effective dielectric constant and fringing effects at the edges.

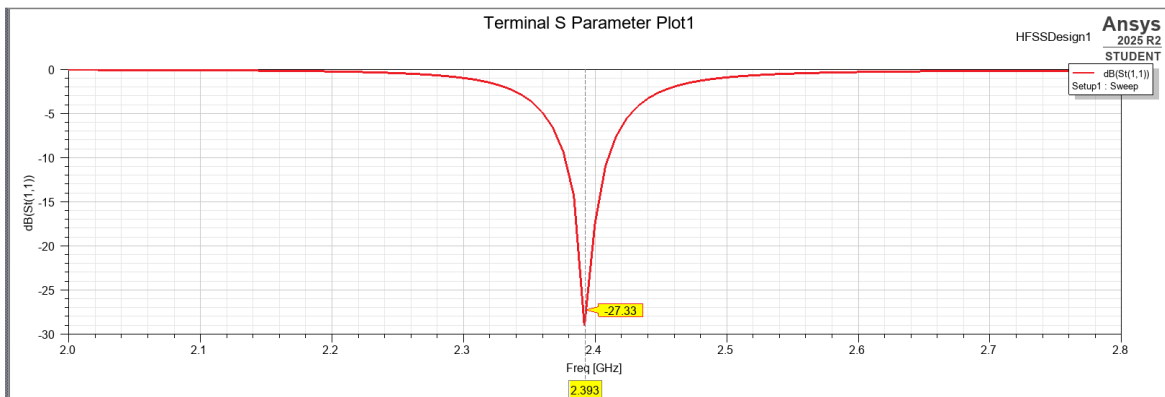


Figure 5.1.a Simulated S- S-parameter of a single element

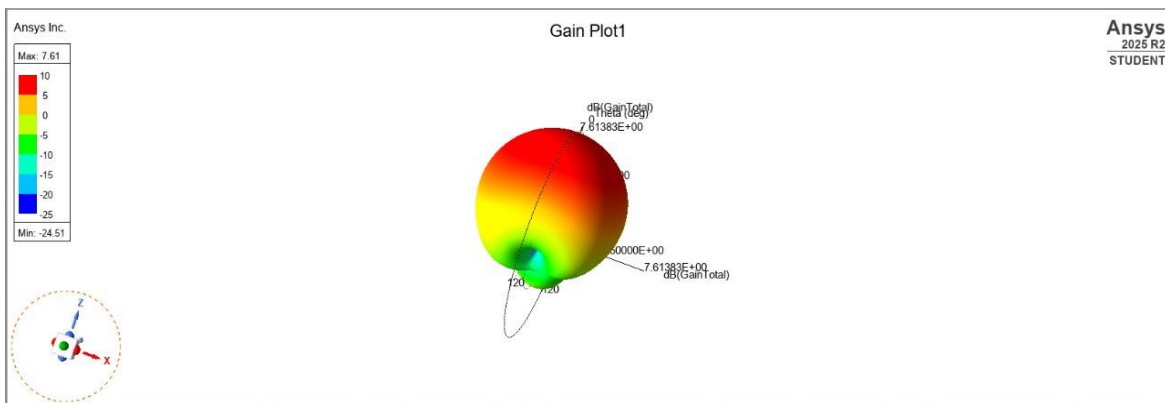


Figure 5.1.b-Simulated Gain plot of a single element

As shown in Table 5-1a, antenna design has sufficiently good input matching (where  $|S_{11}| = -10$  dB) at 2.4 GHz (-27.33 dB).

## 5.2 Performance analysis of Two- element array

The two-element linear array was designed by extending the single rectangular patch antenna described earlier. From a theoretical perspective, a linear array combines the fields of multiple identical elements to enhance gain and directivity. In our case, the two patches are placed side by side along a straight line, with a centre-to-centre spacing of more than  $0.5 \lambda_0$  ( $\approx 62.5$  mm at 2.4 GHz). This spacing is chosen based on array theory to ensure constructive interference in the broadside direction while avoiding significant grating lobes.

Both patches are fed through individual quarter-wave transformers, which are connected to a common  $50 \Omega$  feedline. The transformers ensure proper impedance matching for each element, theoretically transforming the patch input impedance to match the feedline and minimise reflection losses. The simulated  $S_{11}$  shows a strong resonance at 2.4 GHz, with values below  $-20$  dB, confirming that the array is well-matched at the target frequency.

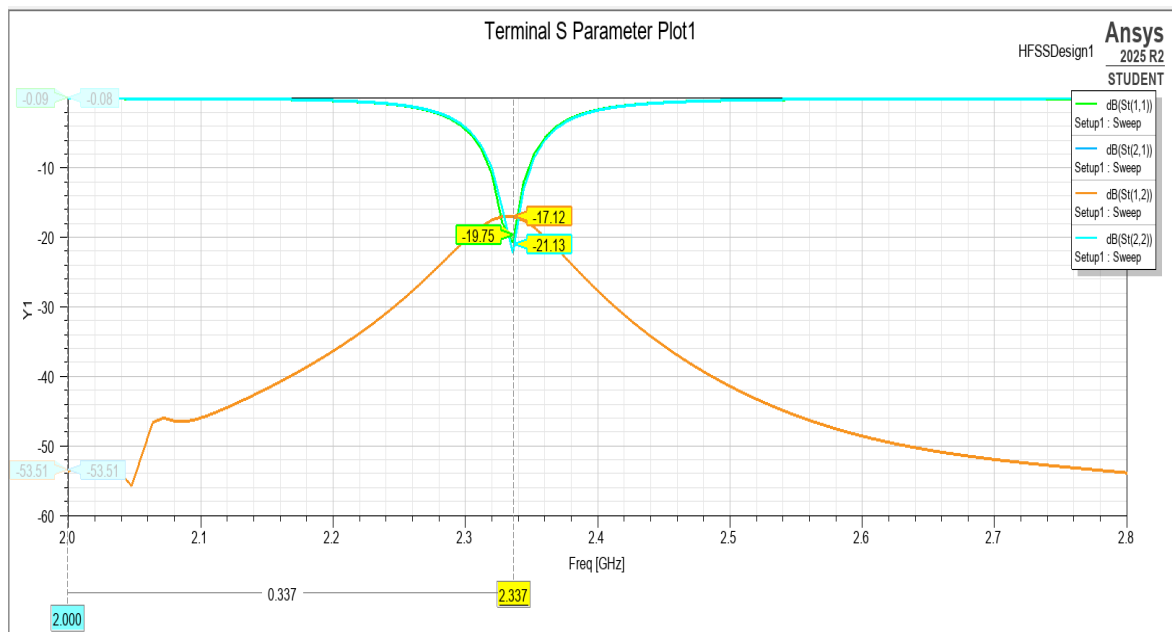
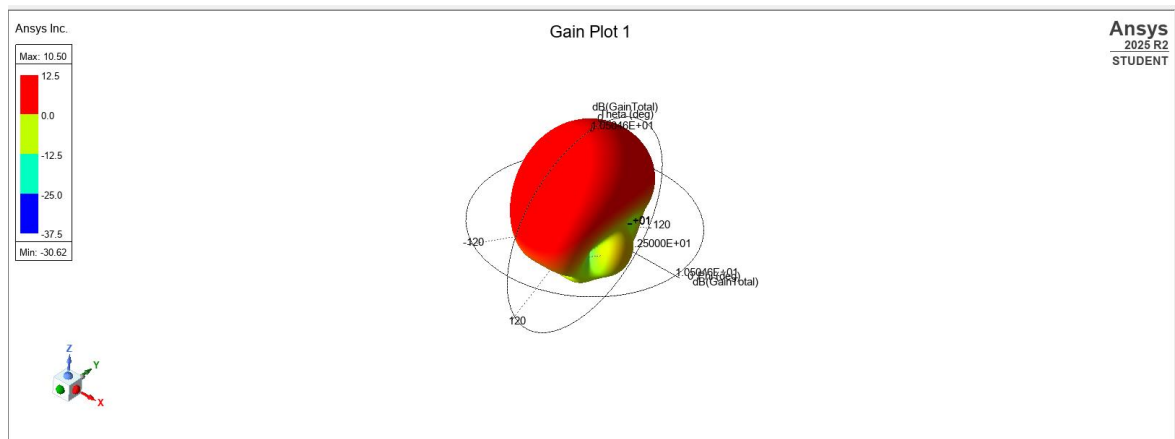


Figure 5.2.a-Simulated S-parameter plot of a single element



## 6. Conclusion and Future Work

The design and analysis of a microstrip patch antenna operating at 2.4 GHz were successfully completed. The single-element antenna was first designed using standard theoretical equations and verified through simulation, showing good return loss, proper impedance matching, and stable radiation at the desired frequency. It exhibited a broadside radiation pattern with adequate gain and efficiency, making it well-suited for wireless communication applications such as Wi-Fi and Bluetooth. The design was then extended into a two-element array configuration, which demonstrated noticeable improvement in gain and directivity compared to the single-element design. This confirmed the theoretical understanding that arraying multiple elements enhances radiation performance through constructive interference. The simulated results validated the effectiveness of the design, both in terms of impedance matching and radiation behaviour, using a Rogers RT/Duroid substrate and quarter-wave transformer feeding. However, when multiple antenna elements are placed close to each other, mutual coupling becomes an important factor that can affect array performance. Mutual coupling can alter the input impedance, radiation pattern, and overall efficiency of the system. Hence, in future work, efforts can be directed towards studying and reducing mutual coupling between elements to achieve better isolation and more stable performance. This can lead to improved signal quality, reduced interference, and higher diversity gain, which are crucial for advanced wireless systems like MIMO and smart antenna applications.

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